Artifact 16 – Identifying Functions

Driving Question: What makes a relation a function, and how do you check it in a set of ordered pairs, a table, a mapping, or a graph?

Example 1

A snack machine has six buttons. The table shows the price, in dollars, of the item behind each button.

Button	1	2	3	4	5	6
Price	1.25	1.50	1.25	2.00	1.50	1.75

Write the relation as ordered pairs, (button, price).
{(1, 1.25), (2, 1.50), (3, 1.25), (4, 2.00), (5, 1.50), (6, 1.75)}

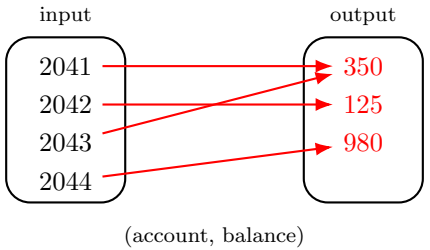
Is the price a function of the button? Explain.
Yes. Every button (input) has exactly one price (output). No button is listed twice.

Two buttons cost 1.25 and two cost 1.50. Does that break anything? Explain.
No. A repeated OUTPUT is allowed. Only a repeated input with two different outputs breaks it.

Read it backwards, (price, button). Is the button a function of the price? Explain.
No. The input 1.25 has two outputs, 1 and 3 (and 1.50 has 2 and 5). Direction matters.

Example 2

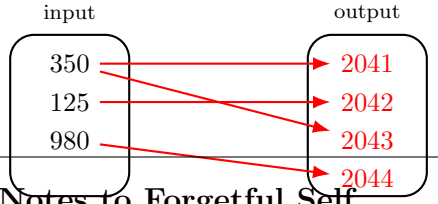
Machine Card 5 from your board: a credit union's four accounts and their balances, (account number, balance in dollars). Draw it as a mapping.



(account, balance)

Is the balance a function of the account number? Explain.
Yes. One arrow leaves every input. 2041 and 2043 both point to 350: two arrows ARRIVING at one output is fine.

In a mapping, what does "not a function" look like?
An input with two or more arrows leaving it.

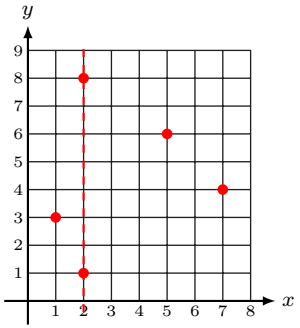


Now reverse it, (balance, account). Sketch that mapping on the left. Is the account a function of the balance? Explain.
No. Two arrows leave 350 (to 2041 and 2043): one input, two outputs. Direction matters.

Example 3

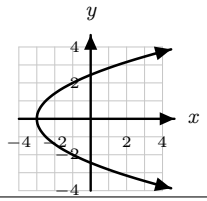
The table shows a relation. Plot its points.

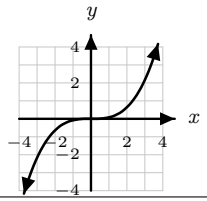
x	1	2	2	5	7
y	3	8	1	6	4



Is y a function of x? Where does the picture show it?
No. The input 2 has two outputs, 8 and 1: two points stacked on the vertical line $x = 2$.

Draw a vertical line through each graph where it would cross the most points. Which graph represents a function? Explain.
Left: a vertical line crosses twice (one x, two y's), not a function. Right: every vertical line crosses once, a function.





Notes to Forgetful Self

Given	Where a repeated INPUT shows	A repeated OUTPUT means
ordered pairs	the same first number in two different pairs	nothing; still a function
a table	the same x twice with different y's	nothing; still a function
a mapping	two arrows leaving one input	arrows arriving together: fine
a graph	two points on one vertical line (vertical line test)	points on one horizontal line: fine

A relation is a function when every _____ input _____ has exactly one _____ output.

Does the order of the pairs matter? Does the direction, (x, y) or (y, x), matter?
Order: no. Direction: yes (Examples 1 and 2 read backwards).